

## EDUCATION

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### PhD – Ocean and Earth Sciences,

October 2022 – Present

University of Southampton, Southampton, UK

- Research Focus: Compound Flood Modelling, Natural Hazards, Disasters, Risk, Vulnerability

### Bachelor of Science – Geographical Sciences,

Graduated: May 2021

#### Co-operative Placement Program

University of British Columbia, Vancouver BC, CAN

- Discipline Focus: Geospatial Sciences, Geomorphology, Environmental Sciences & Policy
- Minor: Geographic Information Science and Geographical Computation

## WORK EXPERIENCE

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### Flood Risk Model Developer

September 2021 – Present

Fathom Global Ltd, Bristol, UK

- Developed large-scale tropical cyclone and coastal compound flood inundation risk models
- Conducted numerical modelling and probabilistic statistical analysis with Python, Matlab, and R

### NASA DEVELOP Research Analyst & Team Lead

June – August 2021

NOAA National Centers for Environmental Information, Asheville NC, US

- Coordinated project tasks for Illinois agricultural drought monitoring and capacity development
- Designed a Python toolset to evaluate the suitability of remotely sensed, modeled, and in-situ soil moisture data products for assessing drought conditions
- Produced soil moisture anomaly and percentile time-series datasets, multidimensional rasters, and statistical products to support decision-makers including the Illinois State Water Survey

### NASA DEVELOP Research Analyst

June – August 2020

NASA Jet Propulsion Laboratory, Pasadena CA, US

- Created permafrost-subsidence maps for use in identifying road and infrastructure risk and vulnerability in Fairbanks, AK to support transportation and resource management
- Processed and analyzed LiDAR and L & C-band SAR (UAVSAR and Sentinel-1) data, and utilized ArcGis-Pro, QGIS, and Python to assess and map permafrost-induced surface deformation
- Presented findings at NASA Applied Science Week 2020 - <https://youtu.be/4npjdqDdJ4M?t=805>

### Transport Canada Environmental Officer

January – June 2020

Transport Canada, Pacific Environmental Team, Vancouver BC, CAN

- Contributed to Environmental Impact Assessment (EIA), contaminated site remediation, and environmental protection projects in British Columbia
- Developed a Project-Approval-Documents (PAD) report used for long term planning
- Authored Environmental Assessment project reports guiding high-level policy meetings
- Drafted annual budgets and processed invoices for contaminated site remediation projects

### Environmental Assessment Project Officer

September – December 2019

BC Environmental Assessment Office, Metal Mining Operations Team, Victoria BC, CAN

- Contributed to coordination and management of operational projects with Assessment Officers
- Analyzed adverse socio-economic and environmental effects of development projects
- Provided technical assistance on consultations with proponents, stakeholders, officials from federal, provincial, and local governments, Indigenous Nations, and the public on the EA process
- Drafted correspondence to inform consultation requirements and participation strategies

**Soil Science Research Assistant**

**May – August 2019**

BC Ministry of Forests, Lands and Natural Resource Operations, Prince George BC, CAN

- Analyzed pre-glacial lake reconstruction in northern BC by integrating LiDAR data and digitization in assessments of the paleo-shoreline landform using Global Mapper
- Generated and formatted stereo LiDAR imagery from point cloud data for 3D vector terrain analysis using DAT/EM Summit Evolution and Global Mapper

**NASA Research Intern in the Hydrology Laboratory/USRA**

**July – September 2018**

NASA Goddard Space Flight Center, Greenbelt MD, US

- Advanced a citizen science-based landslide monitoring project by adding verified content to the Cooperative Open Online Landslide Repository (COOLR) and Global Landslide Catalogue (GLC)
- Designed satellite-based GIS maps to generate landslide susceptibility for trend and change analysis
- Standardized landslide databases using comparative analysis to create a data flow diagram

**Forestry Research Assistant (Field and Laboratory)**

**May – August 2017, May-June 2018**

UBC Faculty of Forestry, Vancouver BC, CAN

- Conducted fieldwork according to standards of the Canadian National Forest Inventory
- Surveyed and photographed field sites and plot boundaries using GPS units and Avenza software
- Evaluated bulk-mass density of soil, wood, and plant samples, and prepared for carbon analysis

**RESEARCH PROJECTS**

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*Earth Observations to Enhance Drought Monitoring in Illinois*

- <https://develop.larc.nasa.gov/2021/summer/IllinoisDisasters.html>

- <https://www.ncei.noaa.gov/news/nasa-develop-ncei-summer-2021-term>

- <https://appliedsciences.nasa.gov/our-impact/news/nasa-interns-develop-disaster-products-midwest>

*Earth Observations to Identify Permafrost Subsidence and Infrastructure Vulnerability in Alaska*

- <https://develop.larc.nasa.gov/2020/summer/AlaskaTI.html>

*Landslide Susceptibility: Comparison of Global and Regional Models*

- <https://jaqubc.wixsite.com/mkd-landslide>

*Geomorphology in Haida Gwaii: Hillslope-Landslide Area Coupling*

- <https://jaqubc.wixsite.com/370haidagwaii>

*Geographically Weighted Regression Analysis of Wildfires in British Columbia*

- <https://jaqubc.wixsite.com/bc-wildfire-qwr>

**HIGHLIGHTS & QUALIFICATIONS**

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- Technical
  - Programming with Python, Matlab, R, JavaScript, and HTML; plus high performance computing
  - GIS and geospatial analysis using ArcGIS/QGIS/Google Earth Engine/Global Mapper
  - Map and web map design using Adobe Illustrator/Mapbox/Leaflet
  - Relational database applications using SQL/MS Access/Excel
  - Processing and analysis of LiDAR point cloud and SAR data
- Policy & Planning:
  - Environmental Assessment policy and application in project management operations
  - First Nation Consultation practices, regulation and Indigenous land/water policy
- Project Management
  - Strong service orientation and effective communicator, technical writer and public speaker
  - Highly successful in coalition building, team collaboration, and fieldwork